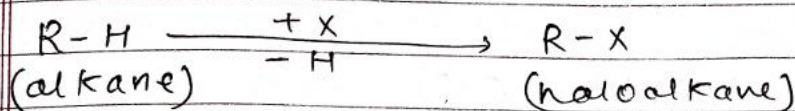


Chapter - I

HALOALKANES

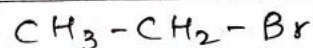
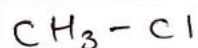


→ Haloalkane is the halogen derivative of alkane obtained by replacing one ^{or more} hydrogen atoms with halogen atoms.

Nomenclature

Common system :- Alkyl halide.

IUPAC " :- haloalkane.

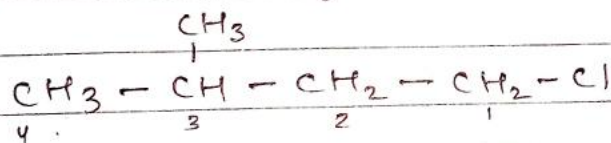


● IUPAC :- Chloromethane

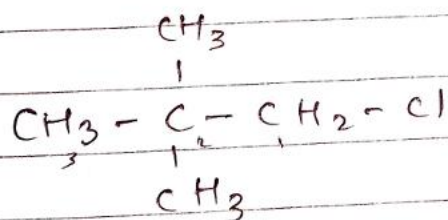
Bromomethane

C.S :- Methyl chloride.

n-Ethyl Bromide.



3-Methyl-1-chlorobutane
iso-pentyl chloride.



~~neo~~ 2,2-dimethyl-1-chloropropane.
neo-pentyl chloride.

Classification of haloalkane.

Based on type of carbon to which halo-group is bonded.

(i) Primary haloalkane (1°)

→ Halogroup bonded with primary ~~group~~ ^{carbon}.
(primary carbon $\Rightarrow$ bonded with one carbon only).

Eg. $\text{CH}_3 - \text{CH}_2 - \text{Cl}$.

(ii) Secondary haloalkane (2°)

→ Halogroup bonded with secondary carbon.
(secondary carbon $\Rightarrow$ bonded with only two carbon).

Eg. $\text{CH}_3 - \underset{\text{Cl}}{\text{CH}_2} - \text{CH}_3$

(iii) Tertiary haloalkane (3°)

→ Halogroup bonded with tertiary carbon.
(Tertiary carbon $\Rightarrow$ bonded with three other carbons)

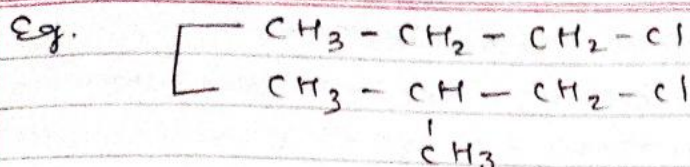
Eg. $\text{CH}_3 - \underset{\text{CH}_3}{\overset{\text{Cl}}{\text{C}}} - \text{CH}_3$

Isomerism

(I) Chain isomerism.

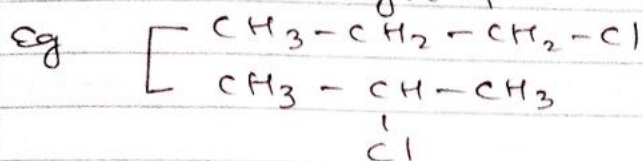
→ Same molecular formula but different length of carbon chain.

$\xrightarrow{h\nu}$



II) position isomerism.

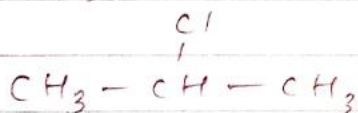
→ Same molecular formula but different position of substituent group.



*) Classification of (R-X) based on no. of halogens in molecules.

i) Monohaloalkanes.

→ Contain one halogen atom.

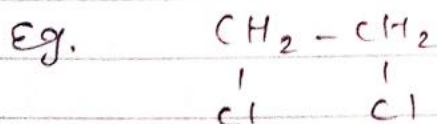


ii) Dihalalkanes.

→ Contain two halogen atom.

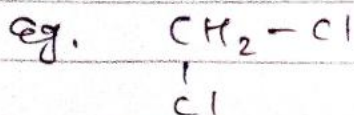
a) Vicinal dihalides

→ Two halogen atom attached to adjacent carbon atom.



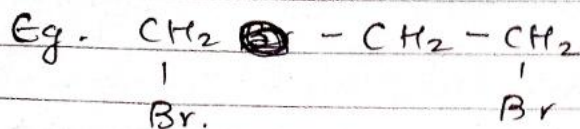
b) Geminal dihalides.

→ Both halogen atoms attached to same carbon atom.



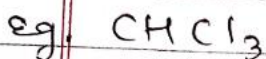
C. polymethylene dihalides:

→ Two halogen atoms are attached to terminal carbon atoms of chain.



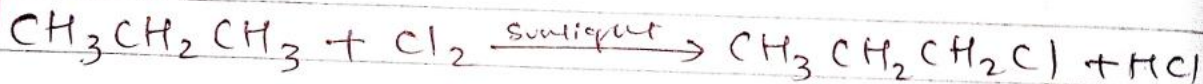
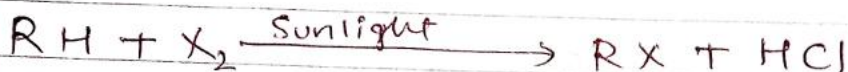
(ii) Polyhaloalkane.

→ Contain more than two halogen atoms.

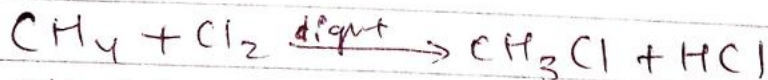


Preparation of Haloalkanes.

(i) from Alkanes.



→ chain rxn:-

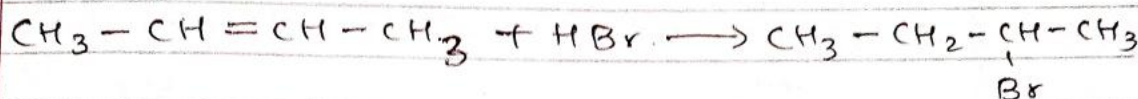


(Chloro form)

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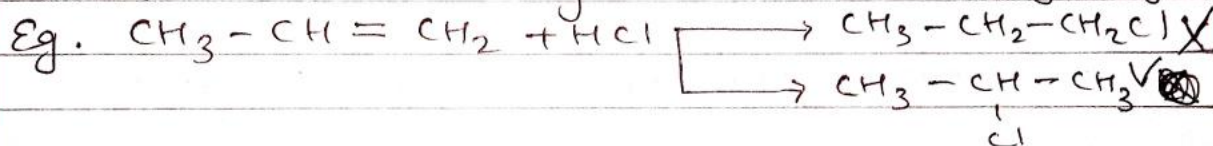
(ii) By reaction between alkene and haloacid / Hydrogen halide.

→ When hydrogen halide is added to alkene, haloalkane is formed.



Markovnikov's rule.

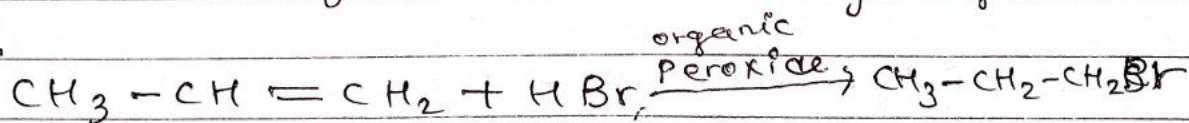
→ When an unsymmetrical reagent is added to unsymmetrical alkene, then the +ve part of reagent goes to that double bonded carbon of alkene which have highest no. of hydrogen and -ve part of reagent goes to other carbon having lowest no. of hydrogen.



Anti-Markovnikov's rule

→ Anti-Markovnikov's rule states that when unsymmetrical reagent is added to unsymmetrical alkene, in presence of organic peroxide, then +ve part of reagent goes with unsaturated carbon bearing minimum no. of hydrogen, and -ve part goes with unsaturated carbon having max^m no. of hydrogen atom.

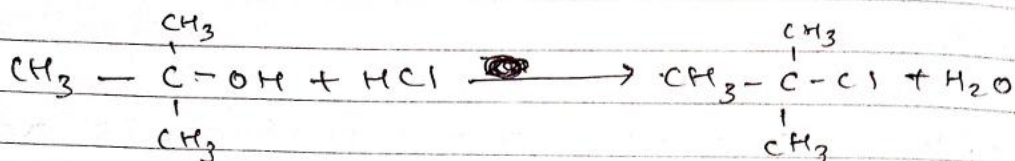
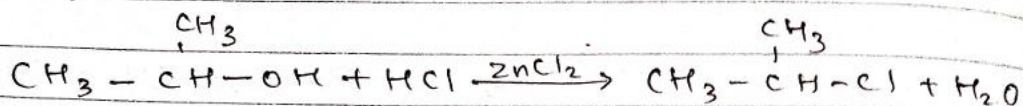
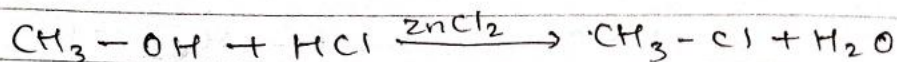
Eg.



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(iii) From alcohol and hydrogen halide.

→ Alcohol when heated with hydrogen halide in presence of ZnCl_2 (Lucas reagent) haloalkane is formed.



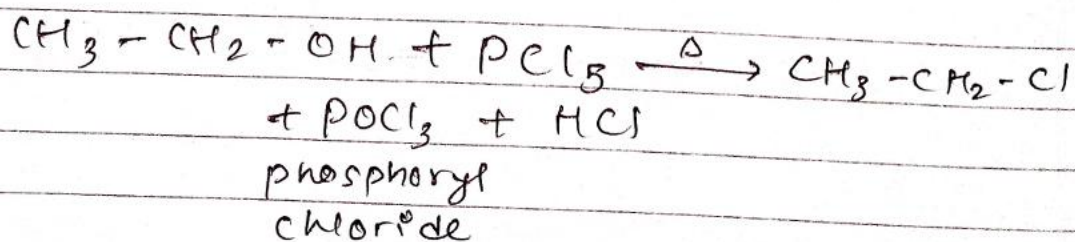
→ Reaction between ~~the~~ 3° alcohol and hydrogen halide doesn't need catalyst because tertiary alcohol is highly reactive.

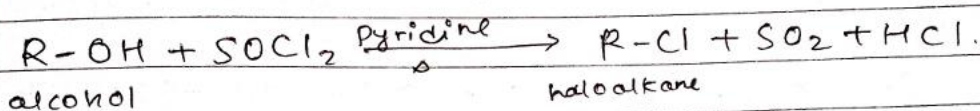
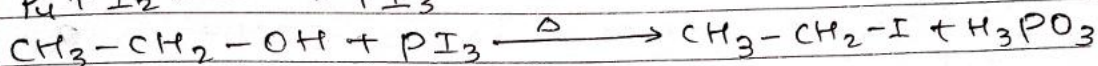
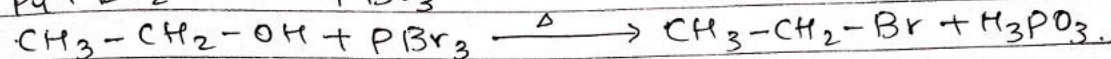


→ Reaction betⁿ alcohol and hydrogen Iodide doesn't need catalyst because hydrogen Iodide is ~~the~~ most reactive haloacid.

(iv) From rxn between alcohol and phosphorus halide or thionyl chloride (SOCl_2)

→ Alcohol when heated with phosphorus halide ~~or~~ thionyl chloride, haloalkane is formed.

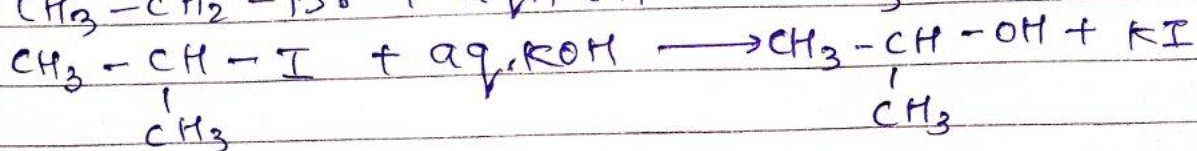
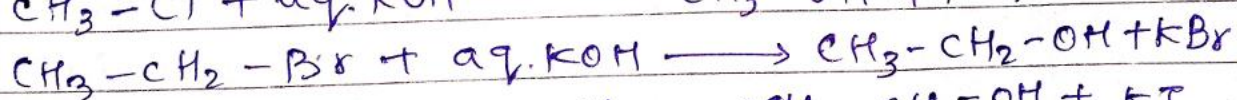
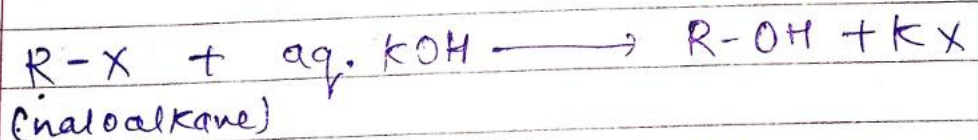




① Nucleophilic substitution rxn.

→ In haloalkane, bond between carbon and halogen is highly polar and easily undergoes dissociation. So haloalkane easily undergoes nucleophilic substitution rxn.

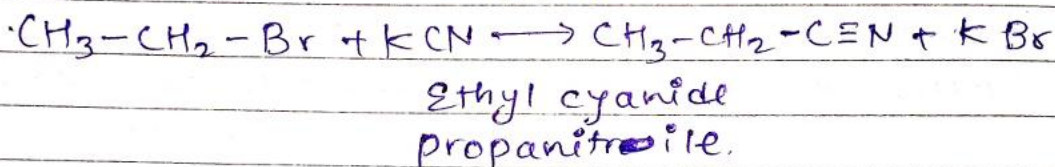
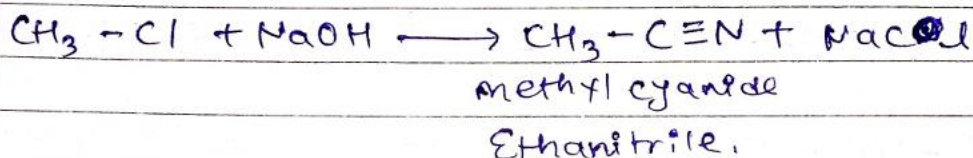
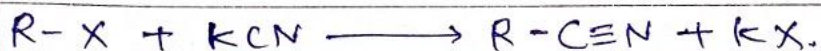
→ Haloalkane when react with aqueous solution of KOH/NaOH , alcohol is formed.



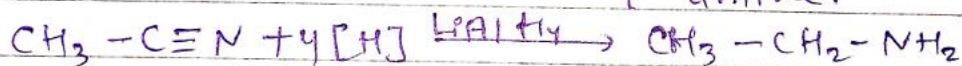
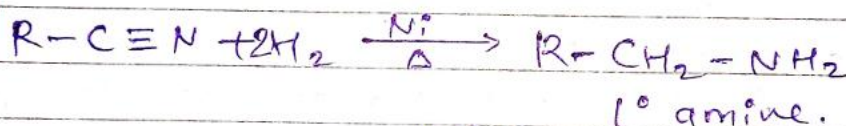
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⑥ Rxn with KCN and NaCN

→ Haloalkane when treated with alcoholic solⁿ of NaCN or KCN, alkyl cyanide is formed.

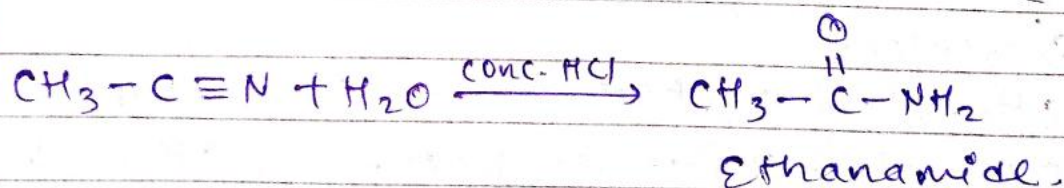
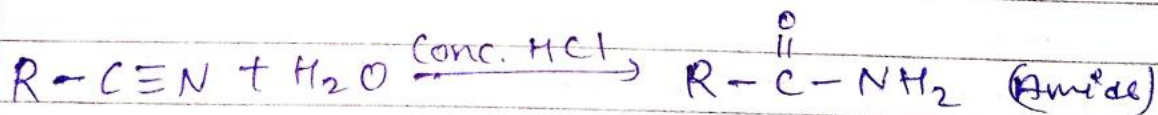


*) Complete reduction of alkyl cyanide gives primary amine.



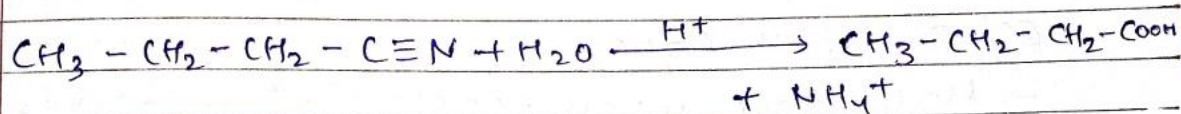
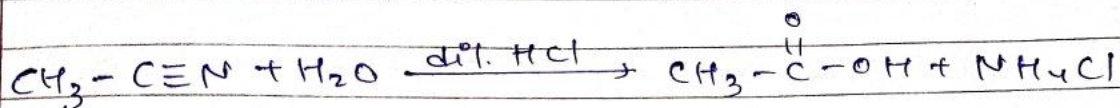
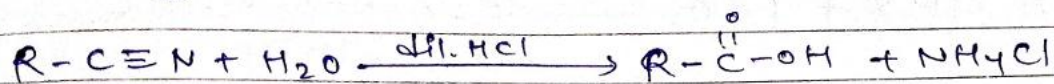
*) Partial hydrolysis of alkyl cyanide.

→ In presence of conc. HCl, alkyl cyanide undergoes partial hydrolysis to give amide.



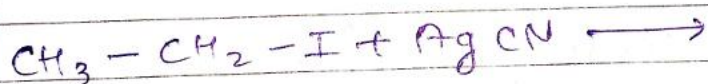
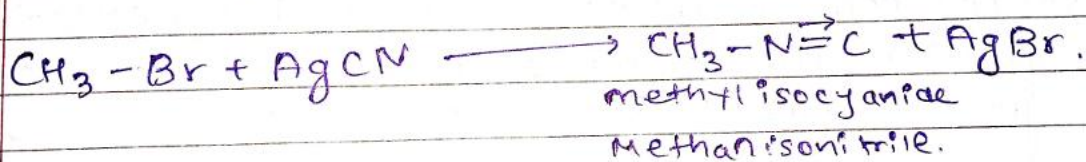
* Complete hydrolysis of alkyl cyanide.

→ In presence of dilute HCl, alkyl cyanide undergoes complete hydrolysis to give Carboxylic acid.

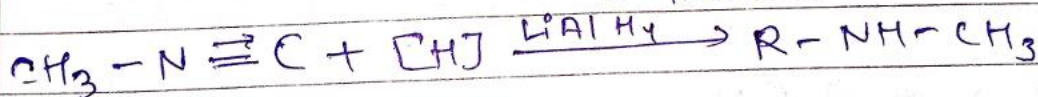
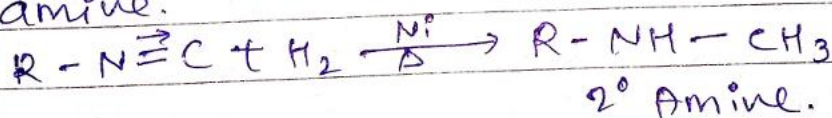


② Rxn with AgCN

→ Haloalkane when react with alcoholic soln of AgCN, isocyanide is formed.



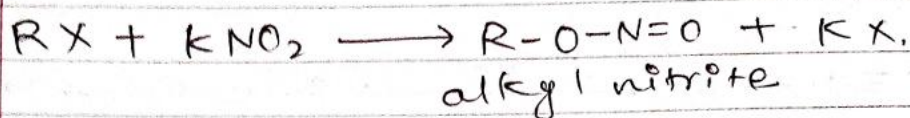
* ③ Reduction of Alkylisocyanide gives 2° amine.



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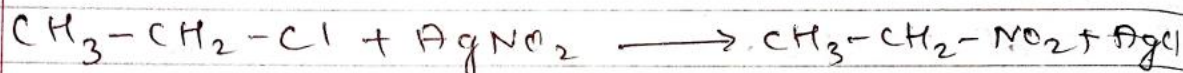
(d) ~~Halo~~ Rxn with KNO_2 or NaNO_2

→ Haloalkane when reacted with aqueous NaNO_2 or KNO_2 , alkyl nitrite is formed.



(e) Rxn with AgNO_2 .

→ Haloalkane when reacted with silver nitrate, nitroalkane is formed.

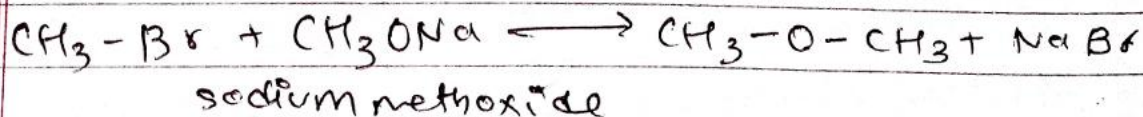
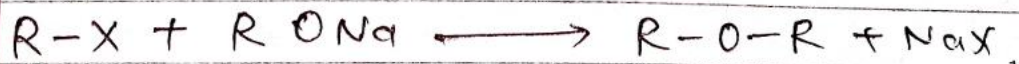


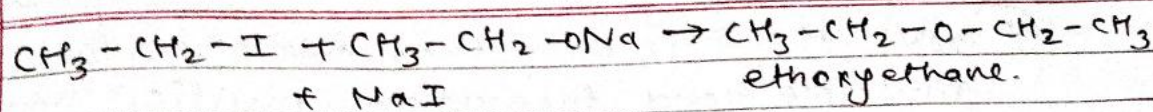
Ambident Nucleophile.

→ Nucleophile which has more than one site, (atom) for bonding is called as ambident nucleophile. Eg: $\text{-C}\equiv\text{N}$, -NO_2

(f) Rxn with sodium alkoxide (Williamson Synthesis)

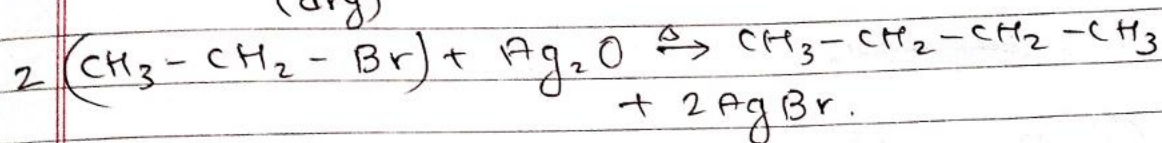
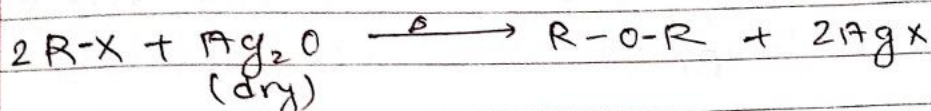
→ Haloalkane when treated with alcoholic solution of sodium alkoxide ether is obtained. This type of rxn is called as williamson synthesis.



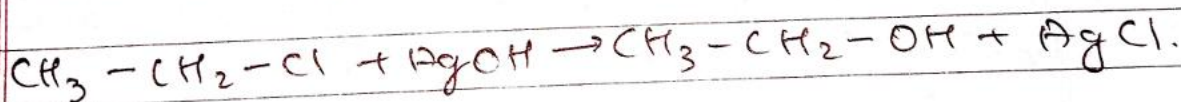
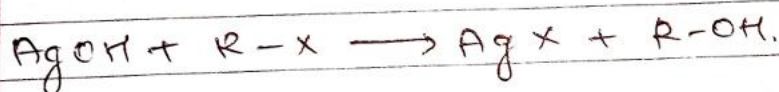
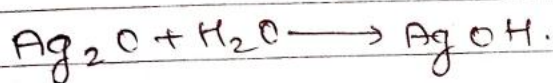


g) Rxn with silver oxide.

→ Haloalkane when heated with dry silver oxide, ether is formed.



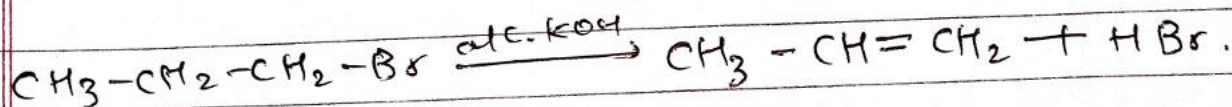
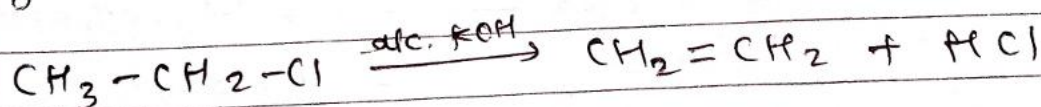
but when moist silver oxide react with haloalkane, alcohol is formed.



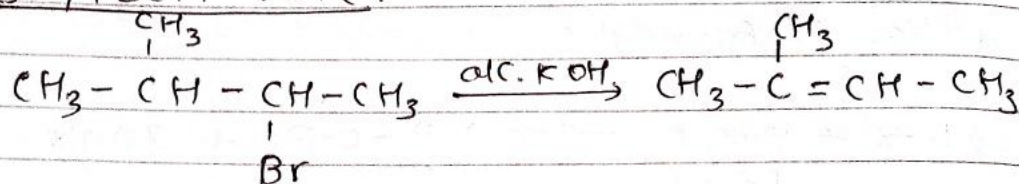
② Elimination rxn.

a) Rxn with alcoholic KOH

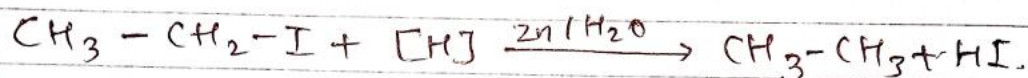
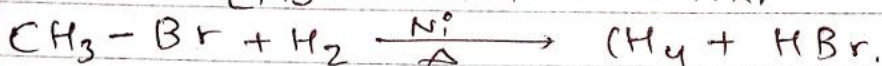
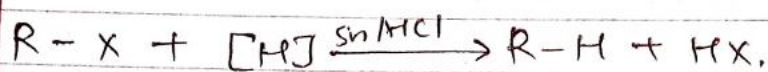
→ when haloalkane is treated with alc. KOH, undergoes dehydrohalogenation to give alkene.



- * When there is more than one side for multiple bond formation, multiple bond goes with most highly substituted carbon atom. This rule is called as Saytzeff rule.

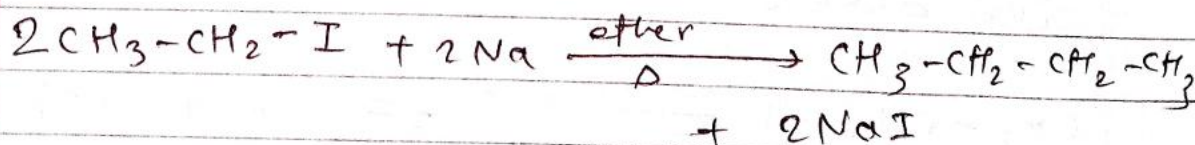
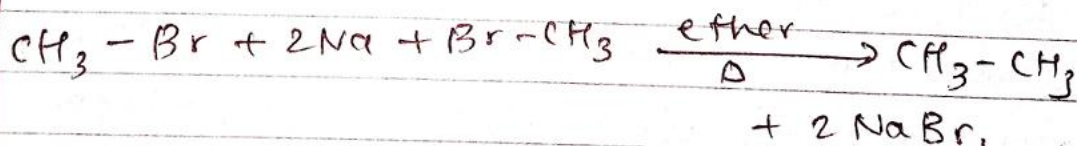
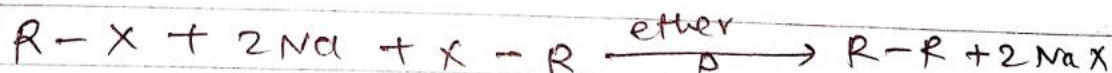


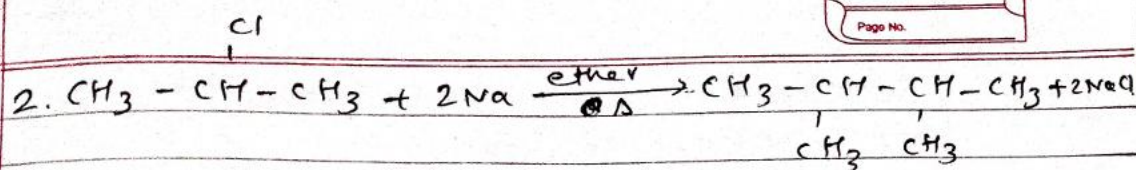
③ Reduction.



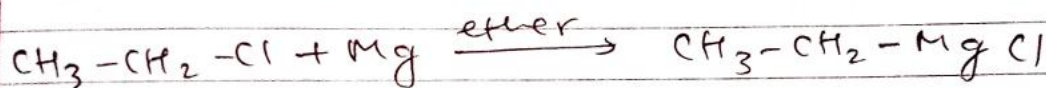
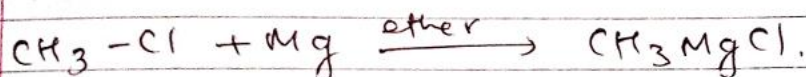
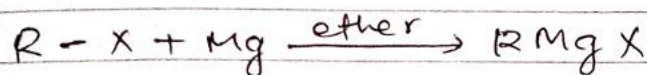
④ Wurtz Rxn.

Halalkane when heated with sodium metal in presence of ether, alkane having double no. of carbon is formed.





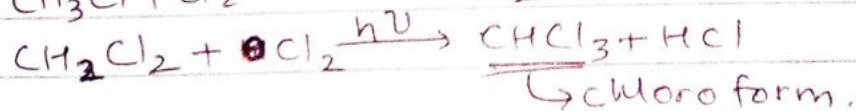
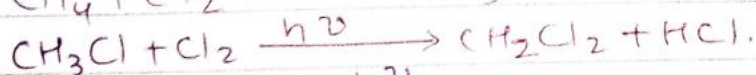
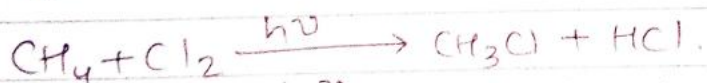
- ⊕ Rxn with magnesium (Grignard rxn).
 → when haloalkane is heated with 'Mg' in presence of ether, alkyl magnesium halide called as Grignard reagent is formed.



- ⊕ Chloroform (CHCl_3)
 IUPAC ⇒ trichloromethane.

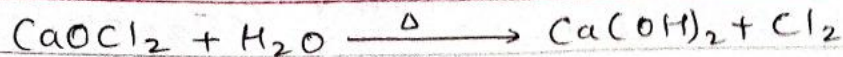
Methods of preparation.

- ① By chlorination of methane.

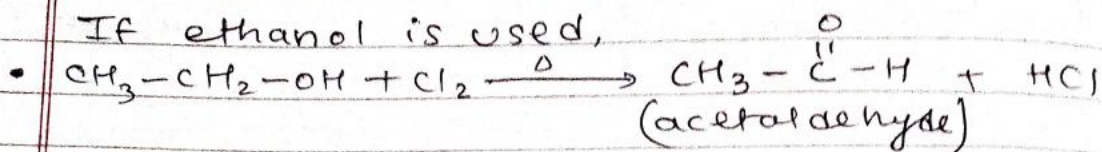


- ② From ethanol or acetone (Lab preparation).

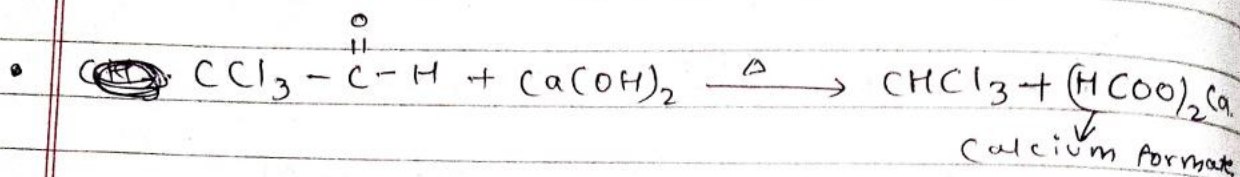
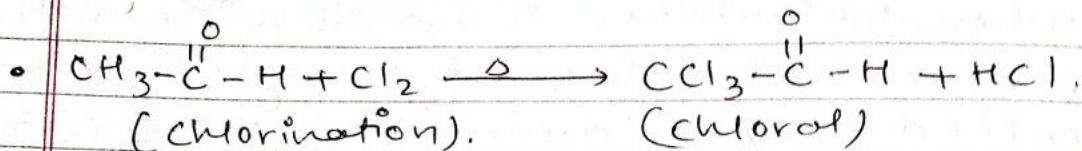
→ Ethanol or acetone when heated with paste of bleaching powder, chloroform is obtained.



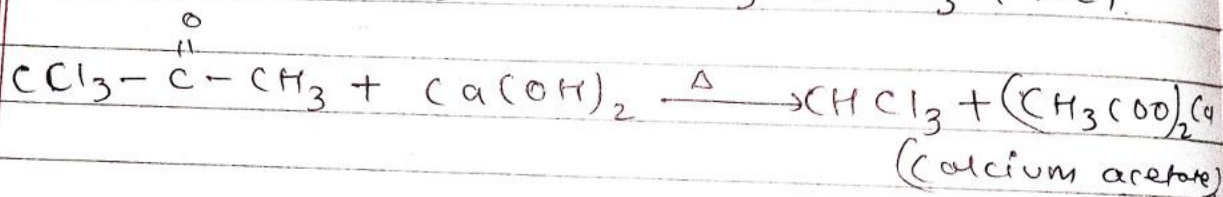
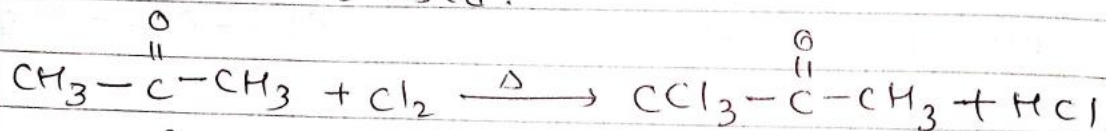
If ethanol is used,



(Oxidation state).



If acetone is used.



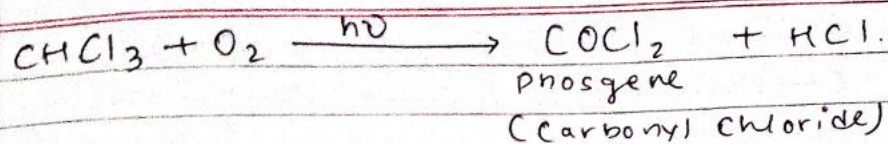
Physical properties of Chloroform.

- Liquid.
- Colourless.
- Sweet odour.
- Partially soluble in H_2O .
- Boiling point = 61°C

Chemical properties.

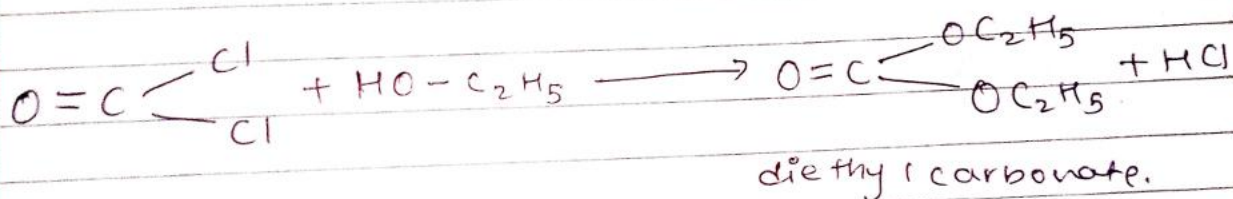
① Oxidation.

- In presence of sunlight, Chloroform is oxidized to poisonous phosgene gas by air.



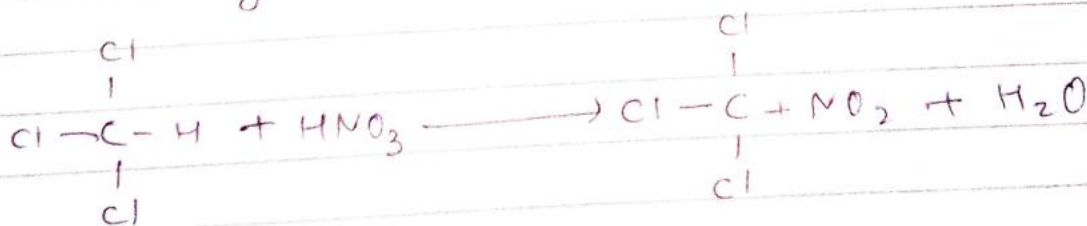
→ In order to reduce chances of oxidation, chloroform is stored in dark brown bottle and completely filled upto the brim.

→ Very small amount of ethanol added to bottle containing chloroform in order to convert poisonous phosgene (if formed) into non-poisonous diethyl carbonate.



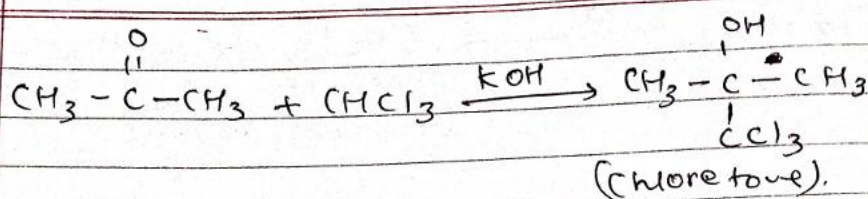
(ii) Rxn with nitric Acid.

→ Chloroform when treated with conc. Nitric acid, chloropicrin is obtained which is used in tear gas.



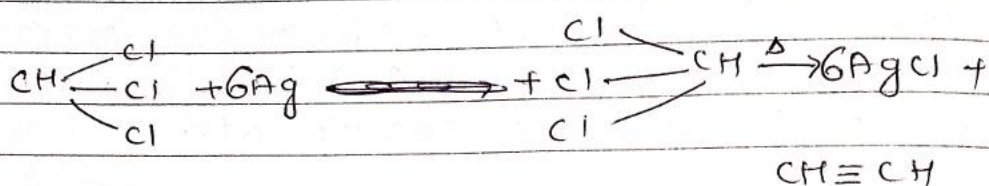
(iii) Rxn with Acetone.

→ Chloroform when react with acetone in presence of KOH, chloroacetone is obtained. which is used as hypnotic drug.

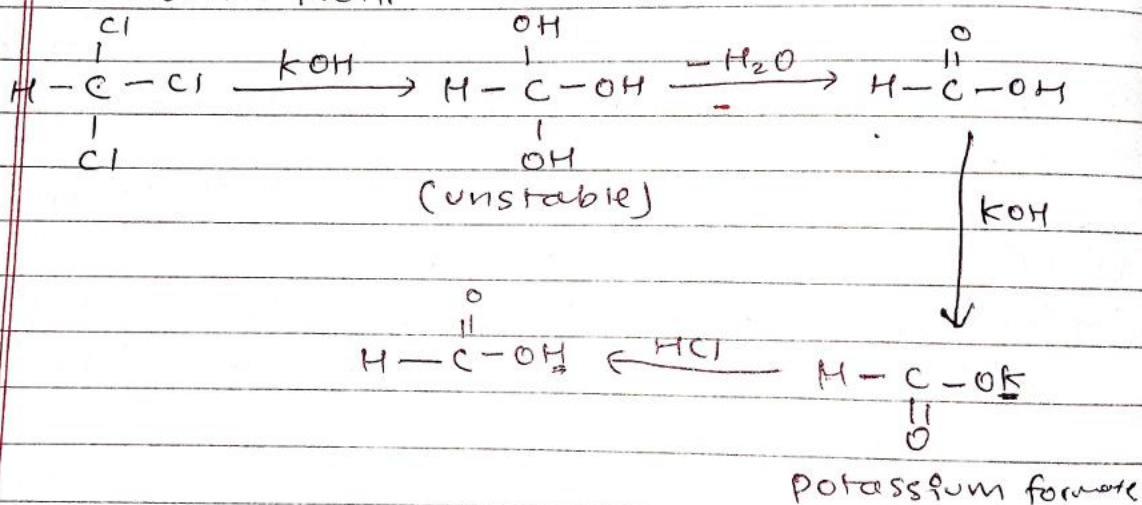


(iv) Rxn with silver powder.

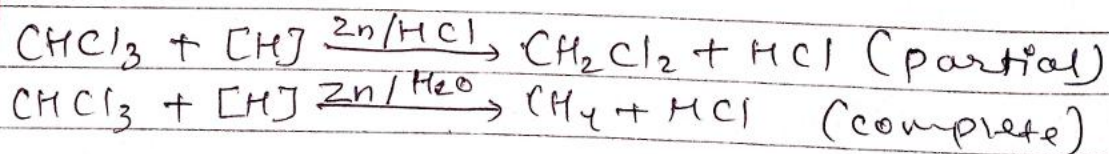
→ Chloroform when heated with silver powder, acetylene is formed.



(v) Rxn with KOH.

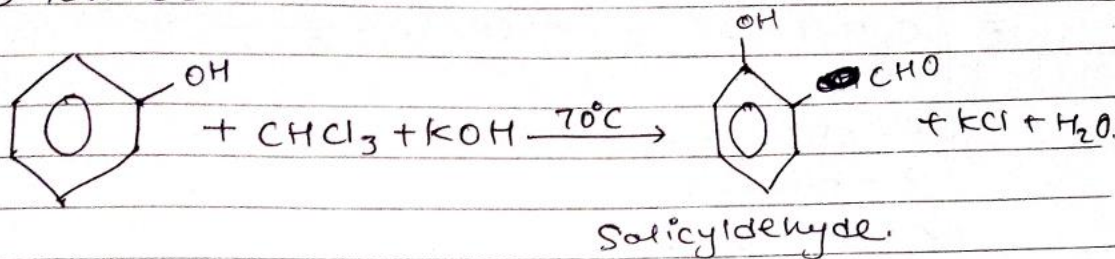


(vi) Reduction.



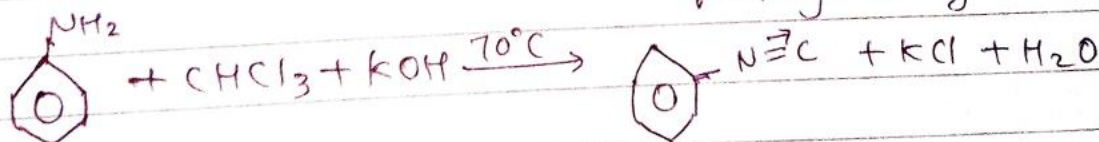
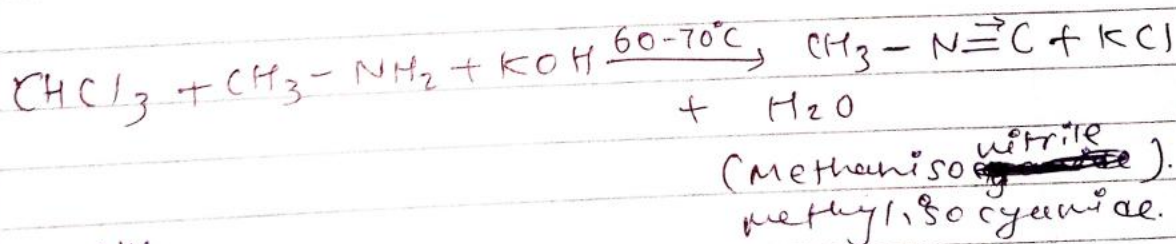
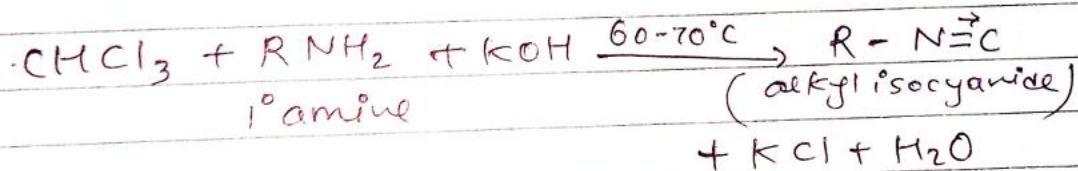
vii) Rxn with phenol [Reimer - Tiemann rxn].

→ Chloroform when heated with phenol in presence of aqueous KOH, ~~the~~ salicylaldehyde is formed.



viii) Rxn with primary Amine. (Carbylamine test)

→ Chloroform when heated with primary amine in presence of alcoholic KOH, offensive odour is obtained due to formation of isocyanide.



Note:- 2° and 3° amines doesn't give ~~the~~ ~~isocyanide~~ ~~test~~
 ⊕ve carbylamine test. So, this test is used to distinguish primary amine from 2°, 3° amine.